RETAIL SALES DATA DASHBOARD

1. Introduction

Business Intelligence (BI) dashboards help organizations analyze data visually and make better business decisions. Retail businesses generate large amounts of sales data every day, and analyzing this data manually is difficult. BI tools such as Power BI help convert raw data into meaningful insights through interactive dashboards and reports.

This project focuses on analyzing retail sales data using Excel, SQL, and Power BI. The project includes data cleaning, SQL analysis, KPI generation, and dashboard development.

The final outcome of this project is an interactive Retail Sales Dashboard that provides insights into sales performance, profits, shipping trends, product performance, and regional analysis.

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2. Objective

The main objectives of this project are:

- Analyze retail sales performance
- Track total sales and profit
- Identify top-performing products
- Evaluate regional sales performance
- Analyze shipping modes and customer segments
- Create interactive visualizations
- Generate business insights for decision-making

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3. Tools and Technologies Used

Tool	Purpose
Excel	Data Cleaning and Transformation
MySQL	Data Import and SQL Analysis
Power BI	Dashboard Development and Visualization

4. Dataset Information

Dataset Source

Kaggle Superstore Retail Dataset

Dataset Description

The dataset contains approximately 9,994 retail sales records of retail sales transactions including customer details, product information, sales, profit, discount, and shipping details.

Important Columns

- Order ID
- Order Date
- Ship Date
- Ship Mode
- Customer Name
- Segment
- Region
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit

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5. Data Cleaning and Transformation

Data cleaning was performed using Microsoft Excel.

Cleaning Steps Performed

5.1 Removed Duplicate Records

Duplicate rows were identified and removed to improve data quality.

5.2 Handled Missing Values

Missing values were checked and corrected wherever necessary.

5.3 Standardized Column Names

Column names were cleaned and standardized for SQL and Power BI compatibility.

5.4 Corrected Data Types

- Date columns were verified
- Numeric columns were formatted properly

5.5 Created Additional Columns

Additional calculated columns were created:

New Column	Description
Year	Extracted from Order Date
Month Name	Extracted month from Order Date
Shipping Days	Difference between Ship Date and Order Date
Profit Margin	Profit divided by Sales

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6. SQL Database Setup

MySQL was used to store and analyze the cleaned dataset.

Database Creation

```
CREATE DATABASE retail_dashboard;
USE retail_dashboard;
```

Table Creation

```
CREATE TABLE superstore_sales (
    row_id INT,
    order_id VARCHAR(50),
    order_date DATE,
    ship_date DATE,
    ship_mode VARCHAR(50),
```

```
    customer_id VARCHAR(50),
    customer_name VARCHAR(100),
    segment VARCHAR(50),
    country VARCHAR(50),
    city VARCHAR(50),
    state VARCHAR(50),
    postal_code VARCHAR(20),
    region VARCHAR(50),
    product_id VARCHAR(50),
    category VARCHAR(50),
    sub_category VARCHAR(50),
    product_name VARCHAR(255),
    sales DECIMAL(10,2),
    quantity INT,
    discount DECIMAL(5,2),
    profit DECIMAL(10,2),
    year INT,
    month_name VARCHAR(20),
    profit_margin DECIMAL(10,4),
    shipping_days INT
);
```

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7. Data Import Process

The cleaned CSV dataset was imported into MySQL using the LOAD DATA INFILE method.

Import Query

```
LOAD DATA INFILE 'cleaned_superstore_utf8.csv'
INTO TABLE superstore_sales
CHARACTER SET utf8mb4
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\r\n'
IGNORE 1 ROWS;
```

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8. SQL Analysis

SQL queries were used to extract business insights.

8.1 Total Sales

```
SELECT ROUND(SUM(sales),2) AS total_sales
FROM superstore_sales;
```

8.2 Total Profit

```
SELECT ROUND(SUM(profit),2) AS total_profit
FROM superstore_sales;
```

8.3 Sales by Region

```
SELECT region,
ROUND(SUM(sales),2) AS total_sales
FROM superstore_sales
GROUP BY region
ORDER BY total_sales DESC;
```

8.4 Top Products by Sales

```
SELECT product_name,
ROUND(SUM(sales),2) AS revenue
FROM superstore_sales
GROUP BY product_name
ORDER BY revenue DESC
LIMIT 10;
```

8.5 Profit by Category

```
SELECT category,
ROUND(SUM(profit),2) AS total_profit
FROM superstore_sales
GROUP BY category
ORDER BY total_profit DESC;
```

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9. Power BI Dashboard Development

Power BI was used to build an interactive dashboard. It helps users understand the data easily through visualizations.

Dashboard Components

Cards

The dashboard includes the following Cards:

- Total Sales
- Total Profit
- Total Orders
- Profit Margin Percentage

Visualizations

Visualization	Purpose
Line Chart	Monthly Sales Trend
Donut Chart	Sales Distribution by Ship Mode
Bar Chart	Most Profitable Categories
Bar Chart	Top 10 Products by Sales

Slicers

Interactive slicers were added for:

- Year
- Region

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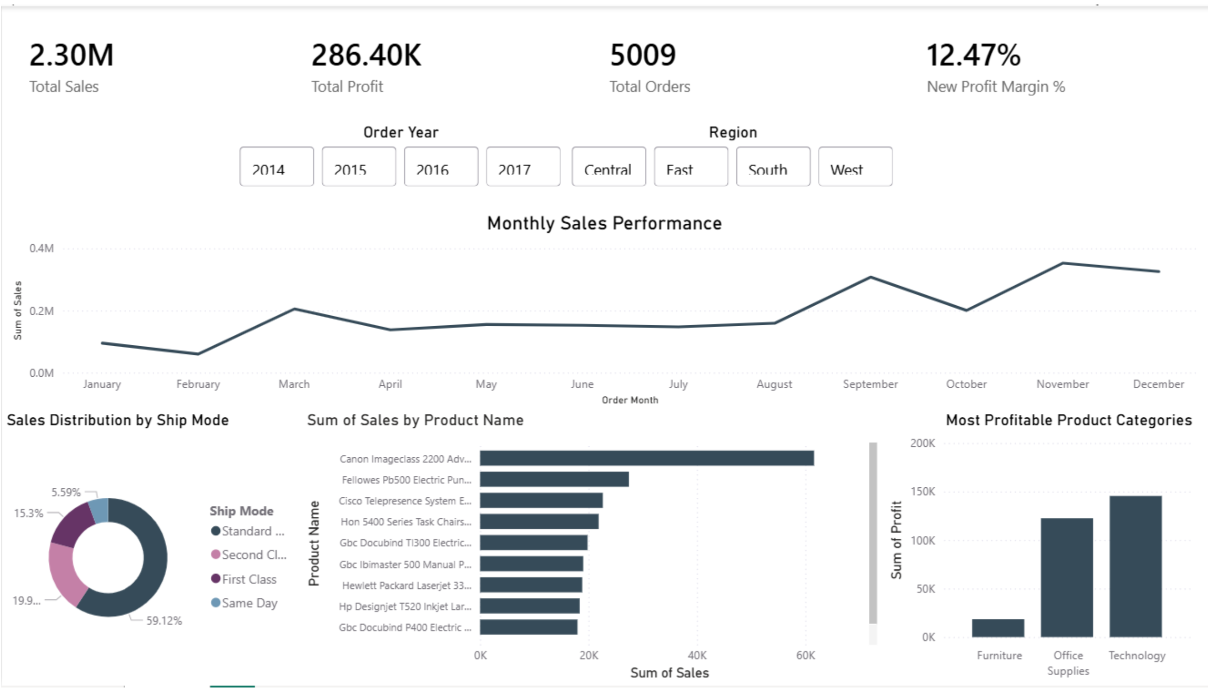
10. Dashboard Features

The dashboard provides:

- Interactive filtering
- Business Cards tracking
- Product performance analysis
- Regional sales analysis
- Shipping mode comparison
- Sales trend visualization



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11. Business Insights

The following business insights were identified:

Insight 1

The year 2017 recorded the highest overall sales and profit, demonstrating significant business growth over time. Meanwhile, 2015 had the lowest sales performance and 2014 generated the lowest profit, indicating the company experienced improved operational and sales efficiency in later years.

Insight 2

Sales performance showed seasonal peaks across different years, with September and November/December consistently generating the highest revenue. This indicates strong year-end and festive season purchasing behavior among customers.

Insight 3

The Canon Imageclass 2200 Advanced Copier was identified as both the highest revenue-generating and most profitable product, making it a key product contributor to the company’s financial success.

Insight 4

The Technology category generated the highest sales and profit among all product categories, establishing it as the company’s most valuable business segment.

Insight 5

The Consumer segment contributed the highest total sales and profit, making it the primary customer segment driving business performance.

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12. Recommendations

The following recommendations identified:

1. Increase marketing and promotional campaigns in low-performing regions such as the South to improve sales growth.

2. Focus on expanding high-performing Technology products due to their strong sales and profitability.

3. Review pricing and discount strategies for Furniture products to improve profit margins.

4. Investigate loss-making products such as the Cubify Cubex 3D Printer to reduce financial losses.

5. Strengthen customer retention strategies for high-value customers such as Sean Miller to maintain long-term revenue generation.

13. Conclusion

This project successfully transformed raw retail sales data into an interactive Business Intelligence dashboard using Excel, SQL, and Power BI.

The dashboard helps users understand business performance, analyze profitability, identify sales trends, and make data-driven decisions.

This project demonstrates practical implementation of Business Intelligence concepts using real-world retail sales data.

The project also improved practical skills in:

- Data Cleaning
- SQL Analysis
- Data Visualization
- Dashboard Development
- Business Intelligence

Submitted By

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